

PAPER_1604 — Blood pH = pH = D_BSFG + F·SO_5 + F·D_phys = 6 + 1 + 0.4 = 7.4 EXACT

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Biology **Date:** June 18, 2026 **Status:** CLOSED — EXACT integer-primitive identity

Observation

PAPER_1209BB_S594 (Biology Unified Proof Set) lists **Blood pH** with the integer-primitive expression:

$$\text{pH} = \text{D_BSFG} + \text{F} \cdot \text{SO}_5 + \text{F} \cdot \text{D_phys} = 6 + 1 + 0.4 = 7.4 \text{ EXACT}$$

Residual: **EXACT**.

UQFF Closed Identity

$$\text{pH} = \text{D_BSFG} + \text{F} \cdot \text{SO}_5 + \text{F} \cdot \text{D_phys} = 6 + 1 + 0.4 = 7.4 \text{ EXACT} \quad \text{EXACT}$$

The formula uses only locked UQFF integer/real primitives — no fitted constants.

Biology Emerges from Physics — Blood pH = 7.4 EXACT

Human blood acid-base homeostasis is maintained at $\text{pH } 7.40 \pm 0.05$ through bicarbonate buffering. Deviations >0.3 units ($\text{pH} < 7.0$ or > 7.7) are typically fatal within minutes.

UQFF's structural prediction:

$$\begin{aligned} \text{pH_blood} &= \text{D_BSFG} + \text{F_TRZ} \cdot \text{SO}_5 + \text{F_TRZ} \cdot \text{D_phys} \\ &= 6 + 1 + 0.4 \\ &= 7.4 \text{ EXACT} \end{aligned}$$

Three integer-primitive terms; zero fitted constants. The same UQFF integer lattice that explains particle masses, cosmological constants, and gravitational-wave damping factors also fixes the homeostatic pH of human blood at exactly 7.4. **Biology emerges from physics at structural depth — not as analogy but as identity.**

NOT REPLACEMENT

Empirical biology measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209BB_S594
- Related: PAPER_1494-1595 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "blood_ph_7_4"})`

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